AGML Lab: Week 1
Linear Regression

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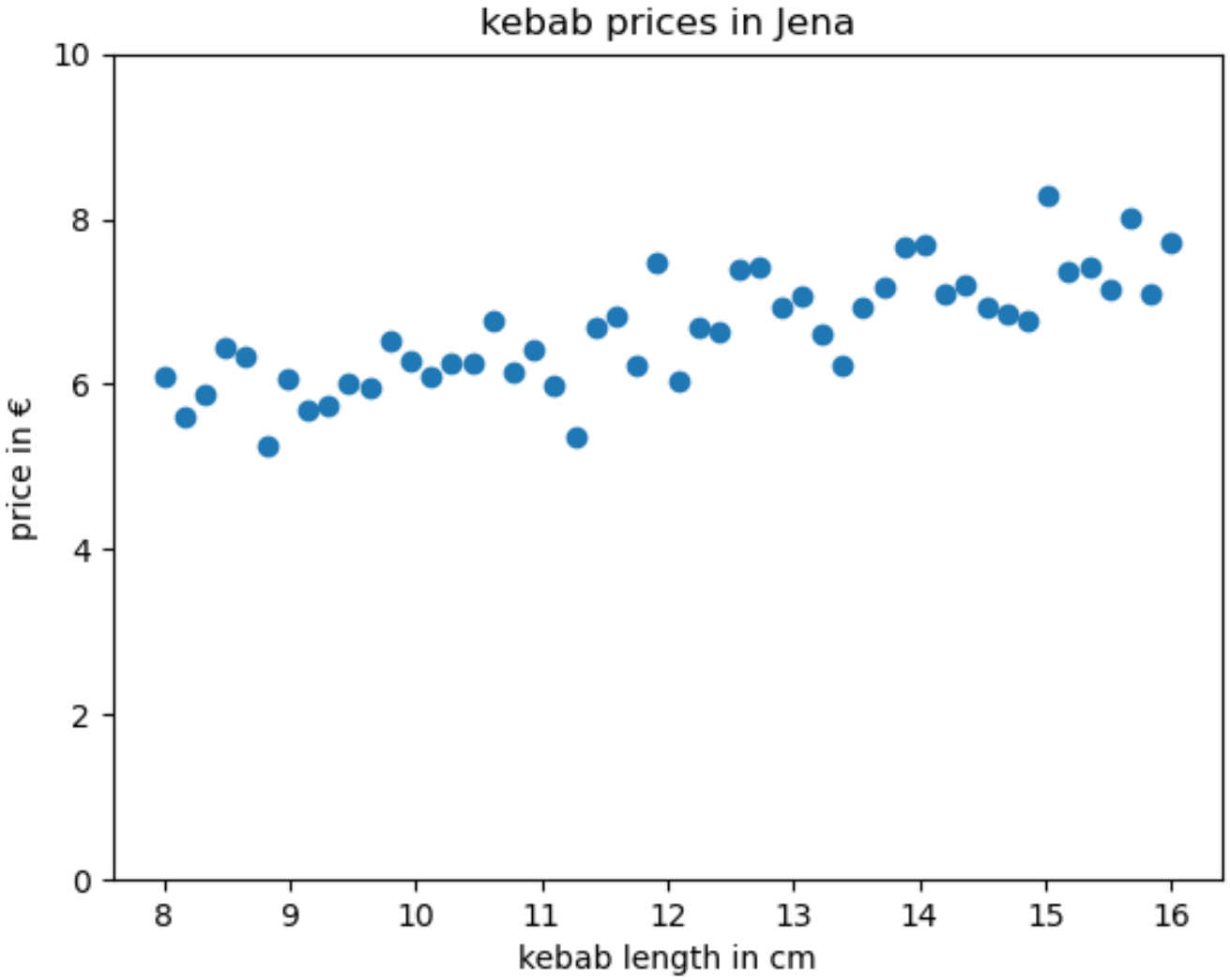
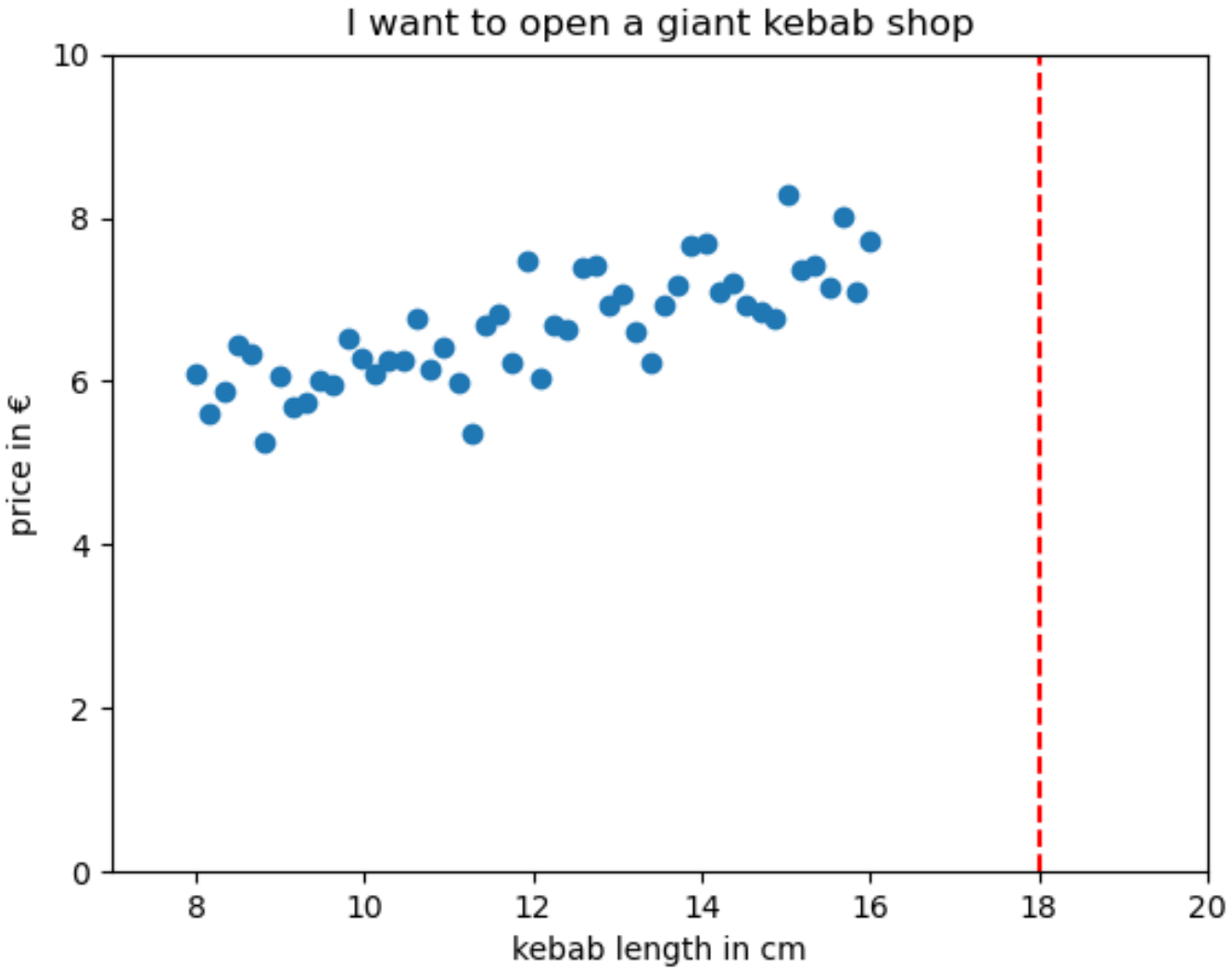
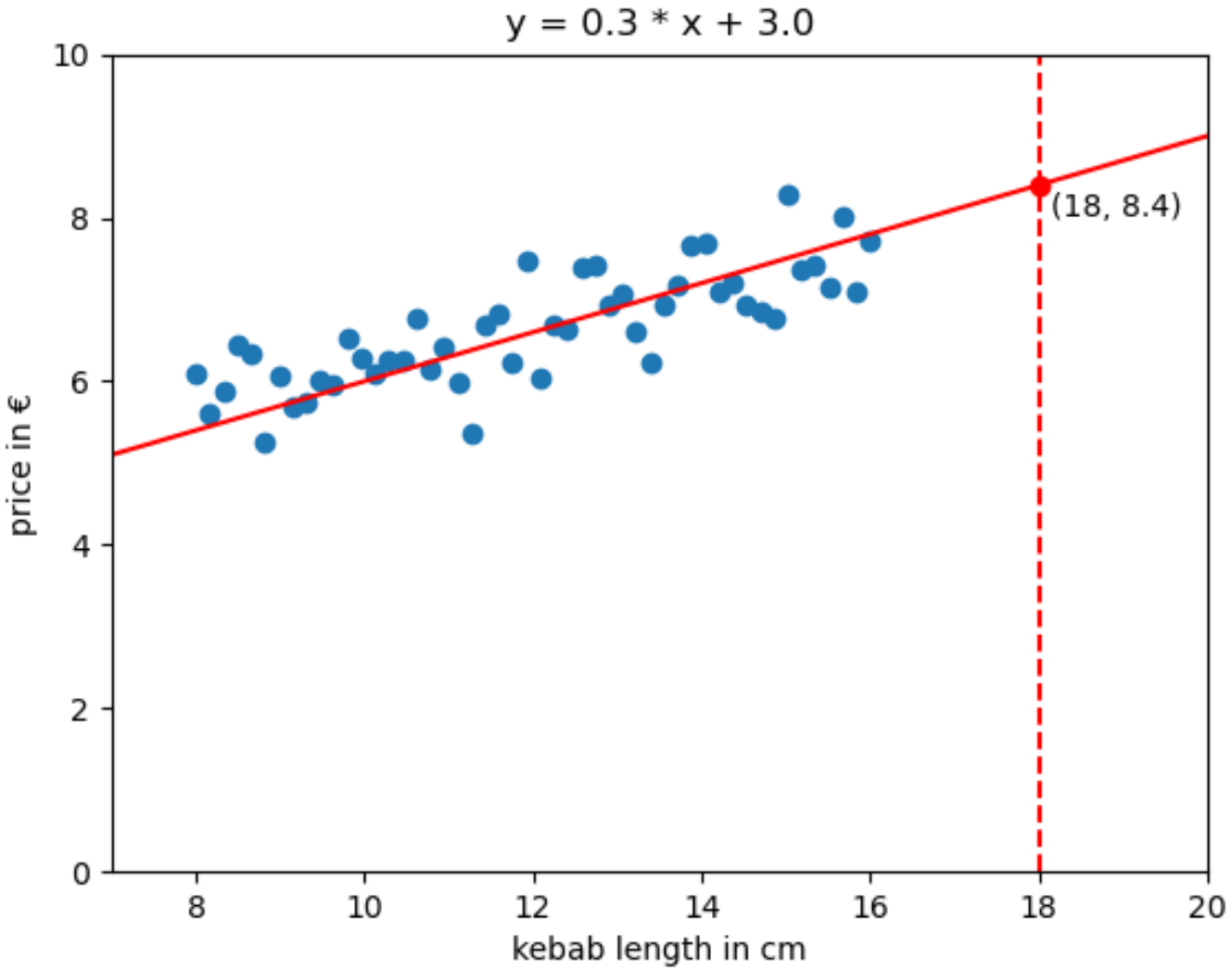
Last Week's Assignment

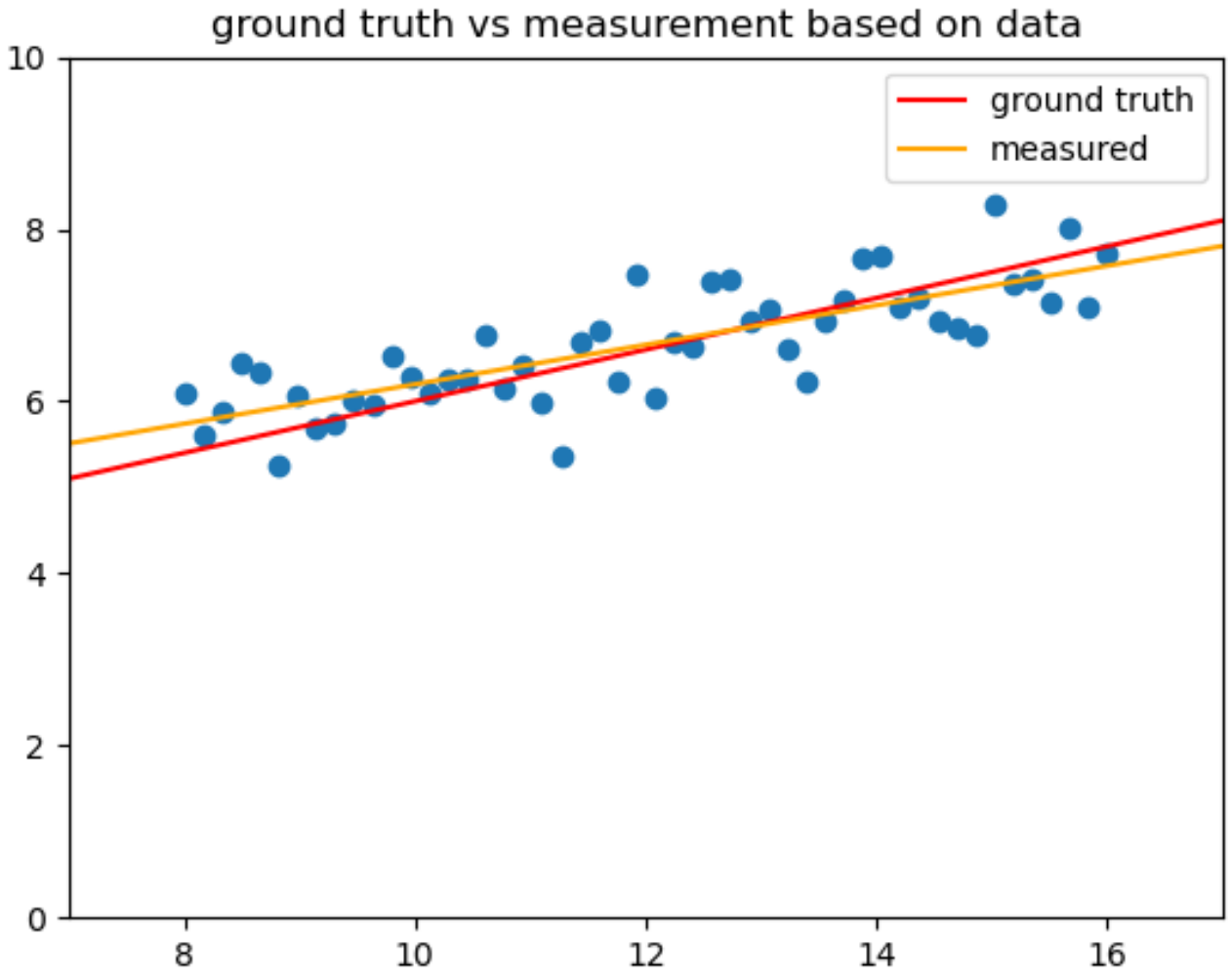
(Jupyter Notebook)

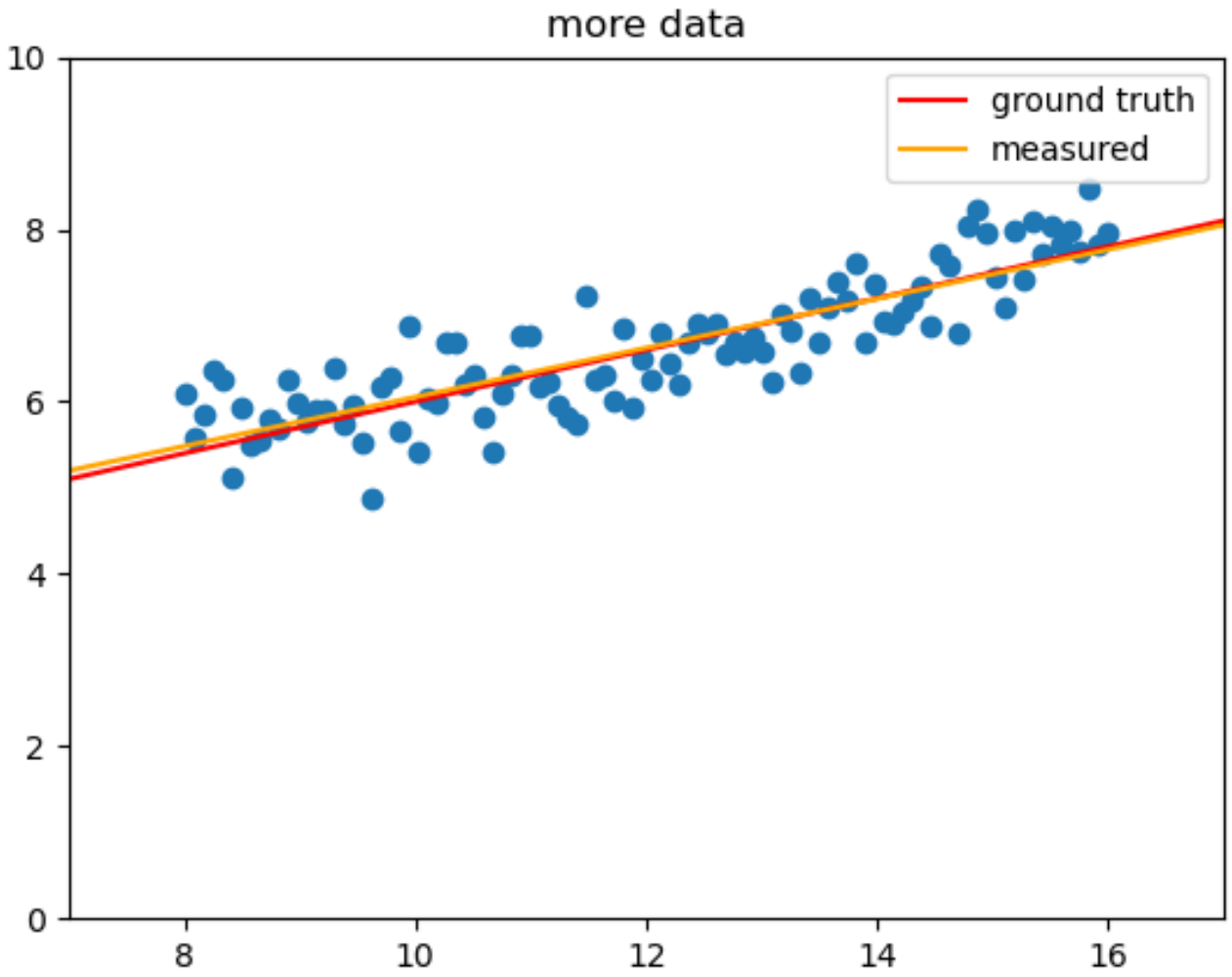
Problem Statement




I want to open a giant kebab shop
price in €
kebab length in cm
0
2
4
6
8
10
8
10
12
14
16
18
20


y = 0.3 * x + 3.0
(18, 8.4)
price in €
kebab length in cm
0
2
4
6
8
10
8
10
12
14
16
18
20


ground truth vs measurement based on data
ground truth
measured
10
8
6
4
2
0
8
10
12
14
16


more data
ground truth
measured
10
8
6
4
2
0
8
10
12
14
16

Model

$$y = X^\top \theta + \varepsilon$$

$$X = \begin{pmatrix} 8 & 8.2 & 8.4 & \dots & 16 \\ 1 & 1 & 1 & \dots & 1 \end{pmatrix}$$

$$\theta = \begin{pmatrix} 0.3 \\ 3 \end{pmatrix}$$

$$y^\top = \begin{pmatrix} 6 & 5.8 & 6.3 & \dots & 7.6 \end{pmatrix}$$

$$\varepsilon \sim \mathcal{N}(0, 0.4)$$

Finding model parameters

Closed-form solution:

$$\hat{\theta} = (XX^\top)^{-1}Xy$$

Caution! Numerically unstable!
Solve instead:

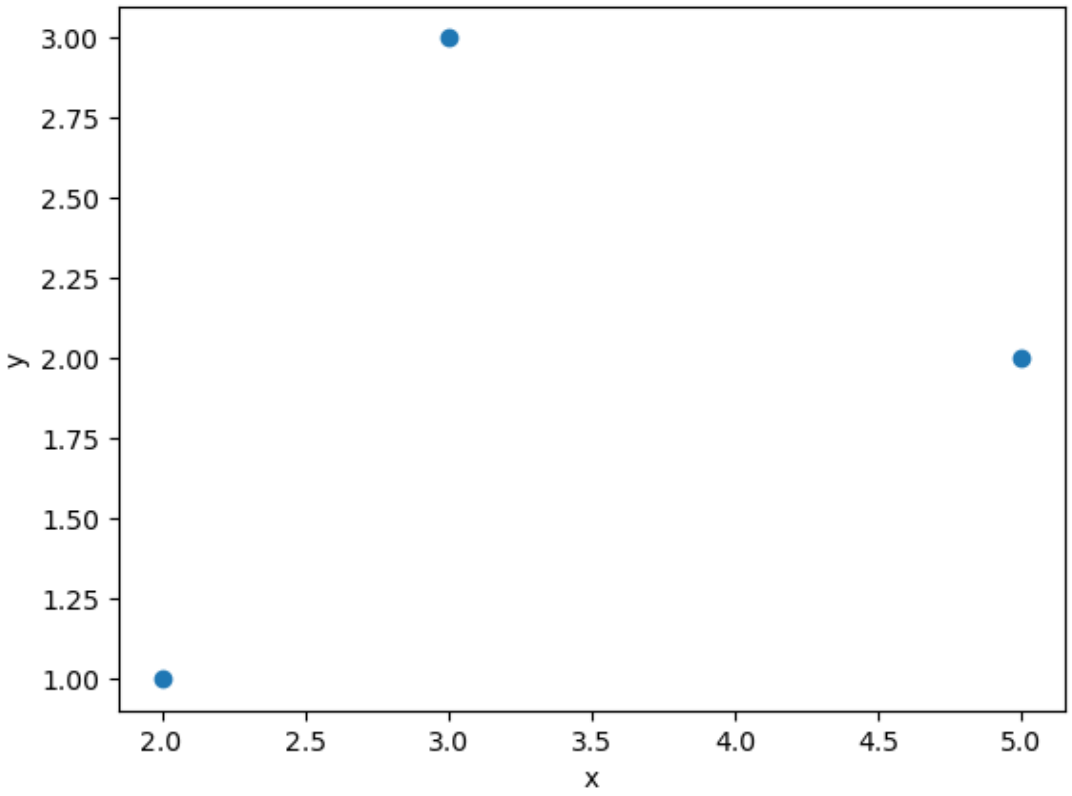
$$XX^\top\hat{\theta} = Xy$$

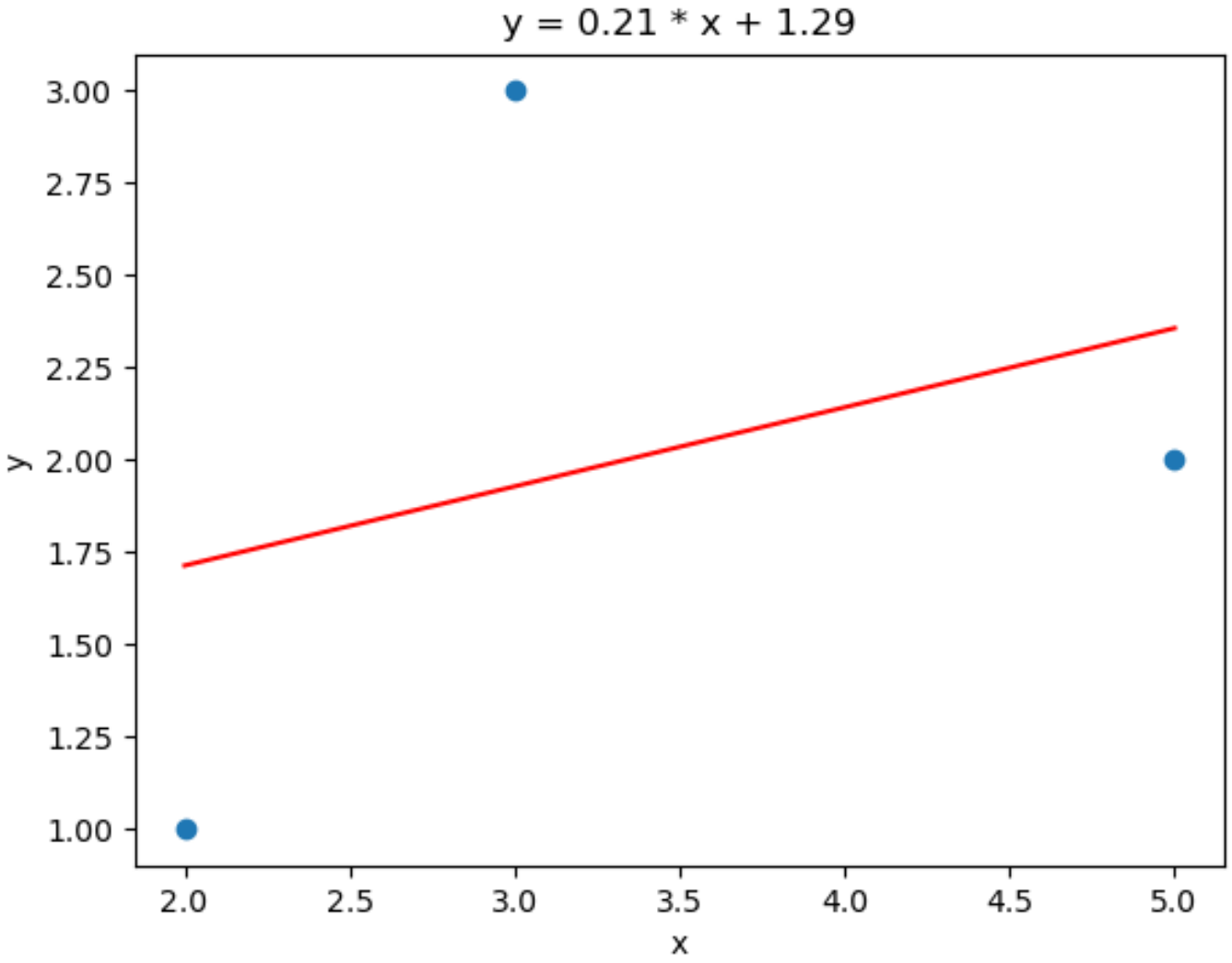
Assignment

(Jupyter Notebook)

Toy Dataset:

$$D = \left\{ \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 3 \end{pmatrix}, \begin{pmatrix} 5 \\ 2 \end{pmatrix} \right\}$$




y = 0.21 * x + 1.29
3.00
2.75
2.50
2.25
2.00
1.75
1.50
1.25
1.00
y
2.0
2.5
3.0
3.5
4.0
4.5
5.0
x

Extra: Derivation of Closed-Form Solution

By modeling our data as $y = X^\top \theta + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ we specifically model the probability of observing a label y_i given the observed sample x_i and a parameter θ:

$$p(y_i | x_i, \theta) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2} \frac{(y_i - x_i^\top \theta)^2}{\sigma^2} \right).$$

Then the following expression tells us: "If this θ was the ground truth, how likely would it be to observe this data?".

$$l(\theta) = \prod_{i=1}^{\#\text{samples}} p(y_i | x_i, \theta)$$

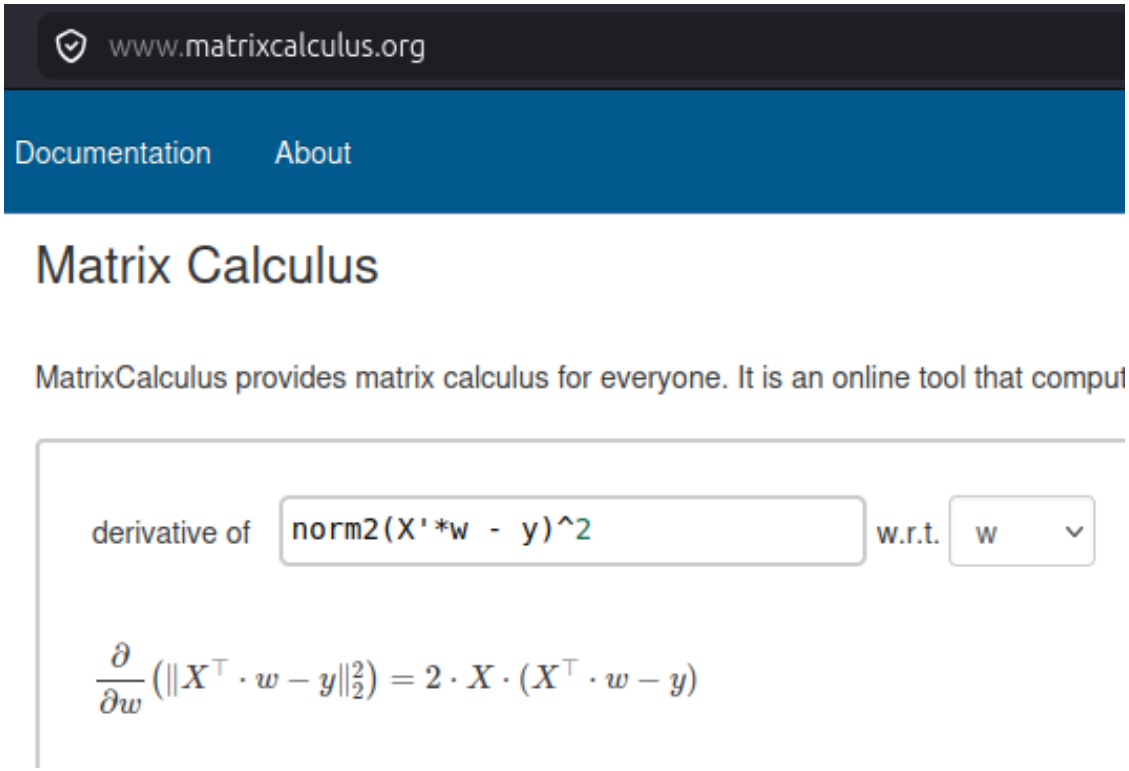
Extra: Derivation of Closed-Form Solution

By maximizing this expression, we find a θ, which maximizes the likelihood of observing the data (Maximum Likelihood Estimate):

$$
\begin{aligned}
\arg\max_{\theta} l(\theta) &= \arg\max_{\theta} \prod_{i=1}^{\#\text{samples}} p(y_i|x_i, \theta) \\
&= \arg\max_{\theta} \sum_{i=1}^{\#\text{samples}} \log p(y_i|x_i, \theta) \\
&= \arg\max_{\theta} -\frac{1}{2\sigma^2} \sum_{i=1}^{\#\text{samples}} (y_i - x_i^\top \theta)^2 \\
&= \arg\min_{\theta} ||X^\top \theta - y||_2^2
\end{aligned}
$$

Extra: Derivation of Closed-Form Solution

How to minimize the loss $L(\theta) = ||X^\top \theta - y||_2^2$?
$\rightarrow$ set the gradient to 0. (get the gradient with help of matrixcalculus.org)



This gives us $X(X^\top \hat{\theta} - y) = 0$, which we can rearrange to get the final solution.

Extra: Ridge Regression

For Ridge Regression, we additionally assume a prior distribution on θ:

$$\theta \sim \mathcal{N}(0, \sigma_\theta^2),$$

which leads to another term in the loss function:

$$L(\theta) = ||X^\top \theta - y||_2^2 + \lambda ||\theta||_2^2,$$

with $\lambda = \frac{\sigma_\varepsilon^2}{\sigma_\theta^2}$, which we can use to control how close to zero we want θ to be.
This has the gradient $X(X^\top \theta - y) + \lambda\theta$, which gives us the new closed-form solution

$$\hat{\theta} = (XX^\top + \lambda \mathbb{1})^{-1} Xy.$$